

Effluent Treatment Plant(ETP)

Part - 2

BIOLOGICAL (SECONDARY) TREATMENT

The principal biological processes used for wastewater treatment can be divided into two main categories:

1. Suspended Growth Process (SGP) i.e., activated sludge Process.
aerated lagoons etc.
2. Attached Growth Process (AGP) i.e., (biofilm) - Trickling filter,
Rotating Biological contactors etc.

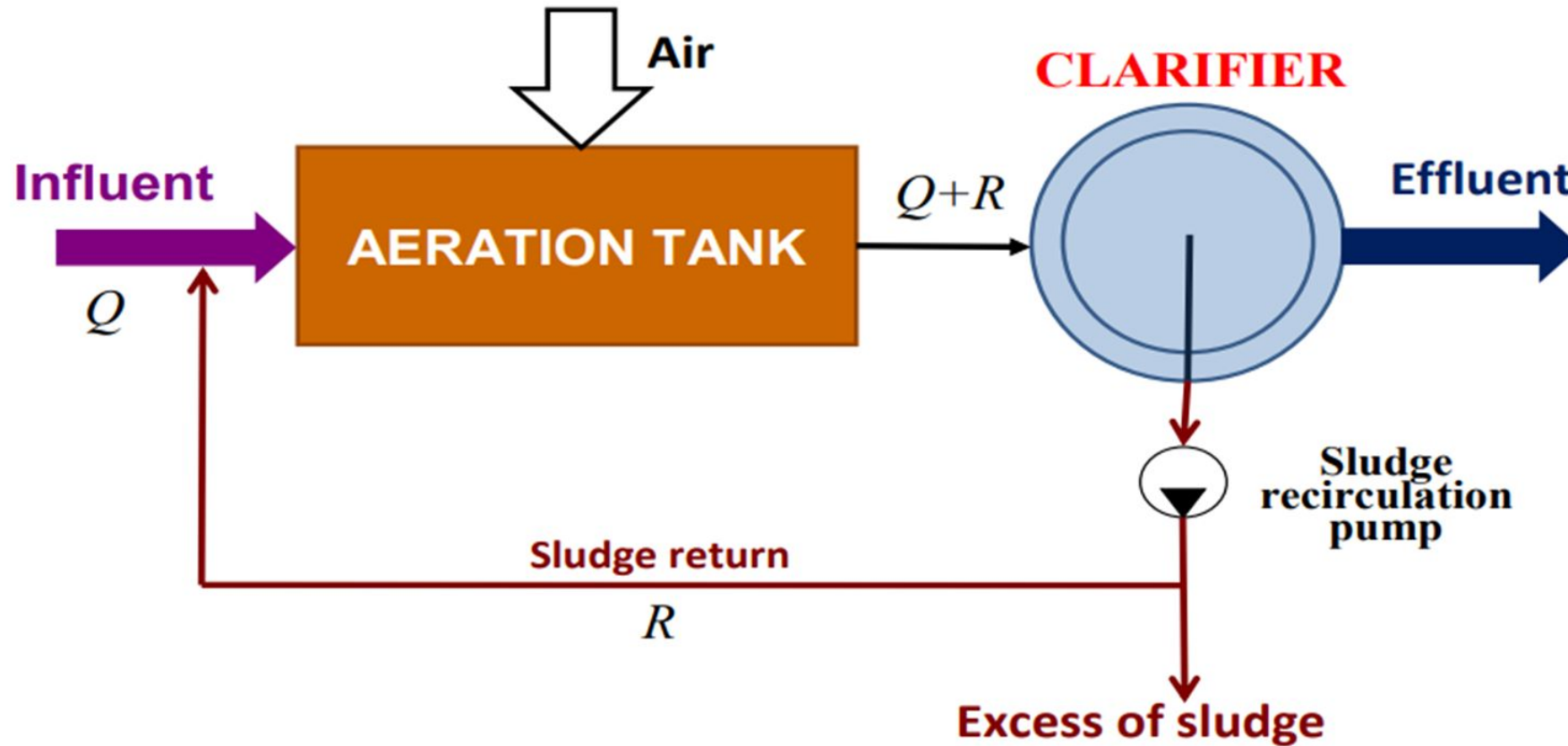
1.(a) Activated Sludge Processes:

In activated sludge processes, the microbes responsible for treatment are maintained in liquid suspension by appropriate mixing methods.

Activated sludge processes are used in both aerobic and anaerobic conditions such as for municipal and industrial wastewater treatment. The activated sludge process can be summarized in the following manner:

Inert matter + organic matter + oxygen + nutrients + microorganisms →
→ → → New microorganisms + CO₂ + H₂O + N₂ + additional inert matter.

Figure: A simplified flow diagram of the activated sludge process:



Microorganisms in activated sludge process:

- The microorganisms are stimulated to convert (eat & digest) harmful, oxygen demanding organic compounds into environmentally more acceptable form (microorganisms) and low energy, stable compounds like water, carbon dioxide and nitrogen gas.
- The microbial community that does that job comprises various species of (I) bacteria, (II) fungi, (III) protozoa, (IV) rotifers (multicellular animals only found in activated sludge), (V) nematodes (round worms) and many other microorganisms.

A microscopic image of activated sludge, showing various types of microorganisms including ciliates, flagellates, and bacteria. The background is a blueish-grey color with many small, dark, irregular shapes representing the sludge flocs and individual organisms.

F/M Ratios

Wasting Rates & SRT

MLSS Concentration

pH

Microorganisms to purify wastewater:



How the microbes work to clean the wastewater:



Some Important Terminology:

- **Biochemical Oxygen Demand (BOD):** is a measure of the amount of oxygen required to decompose organic matter from water by aerobic bacteria or other microorganism.
- **Chemical Oxygen Demand (COD):** in wastewater refers to a measurement of the amount of oxygen needed to chemically oxidize organic matter present in a water sample.
- **Dissolved Oxygen (DO):** DO is the amount of Oxygen gas(O_2) that is dissolved in water.

Aeration Tank in Biological Treatment Process

An aeration tank in wastewater treatment is a reaction tank where wastewater is mixed with air, allowing aerobic bacteria to thrive and break down organic pollutants within the water.

The usual shape of an aeration tank in biological treatment process are rectangular, circular, and oval shaped according to the treatment methods.

Rectangular Aeration Process



Oxidation ditch model:

- The best system for biological treatment of tannery effluents is the **oxidation ditch**, a racetrack-shaped aeration basin, with rotary brush or vertical rotors (carrousel) aerators that extend across the width of the ditch as shown in the following figure.
- Oxidation ditch system has two regimes:
 - (a) **Aerobic regime**: Where oxygen is available in the system consequently the effluent is treated with aerobic bacteria.
 - (b) **Anaerobic regime**: A system, lack of oxygen, where anaerobic bacteria is grown. These bacteria biodegrade some special types of chemical waste.

Oxidation ditch

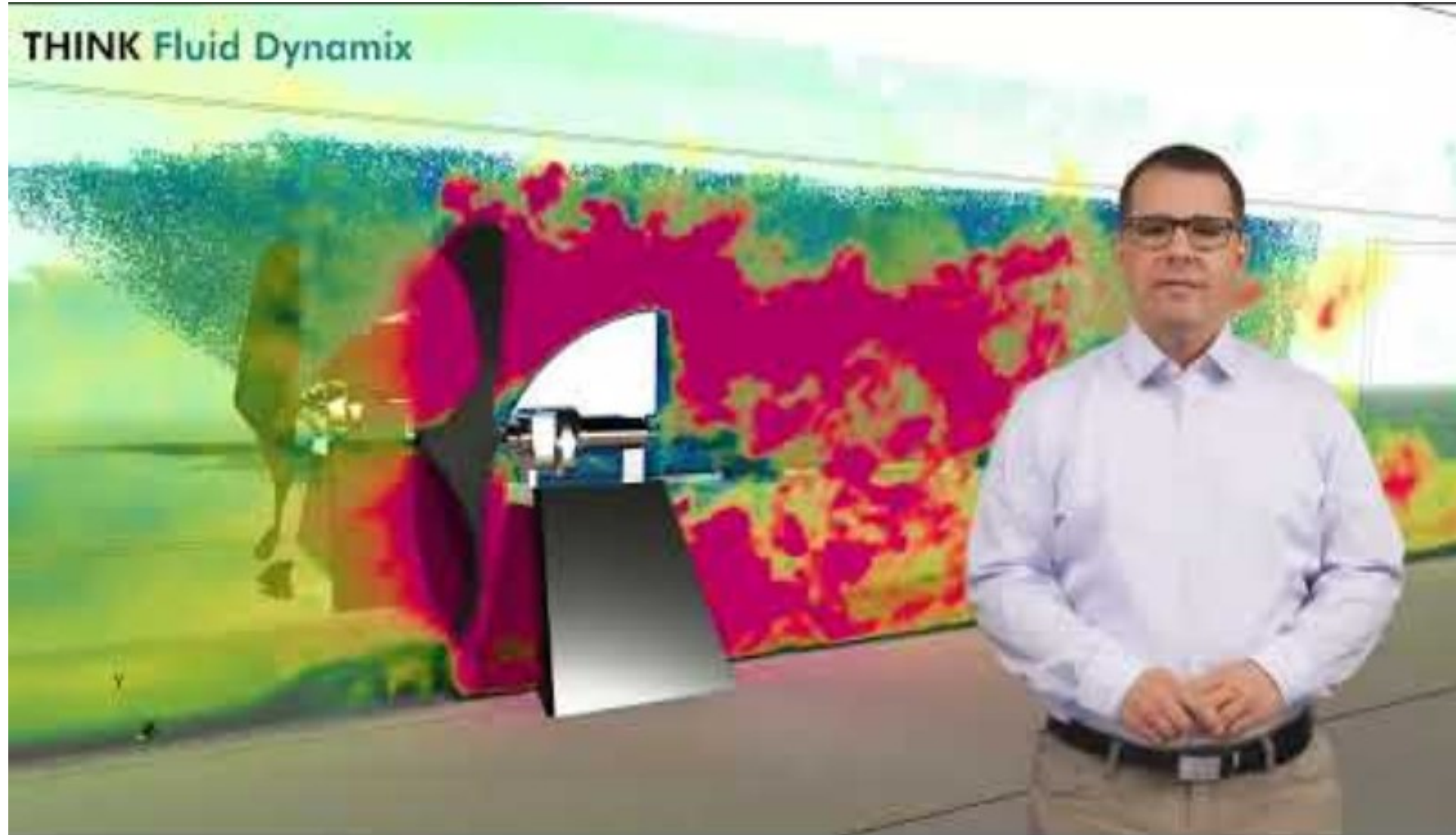


25. View of an oxidation ditch in operation

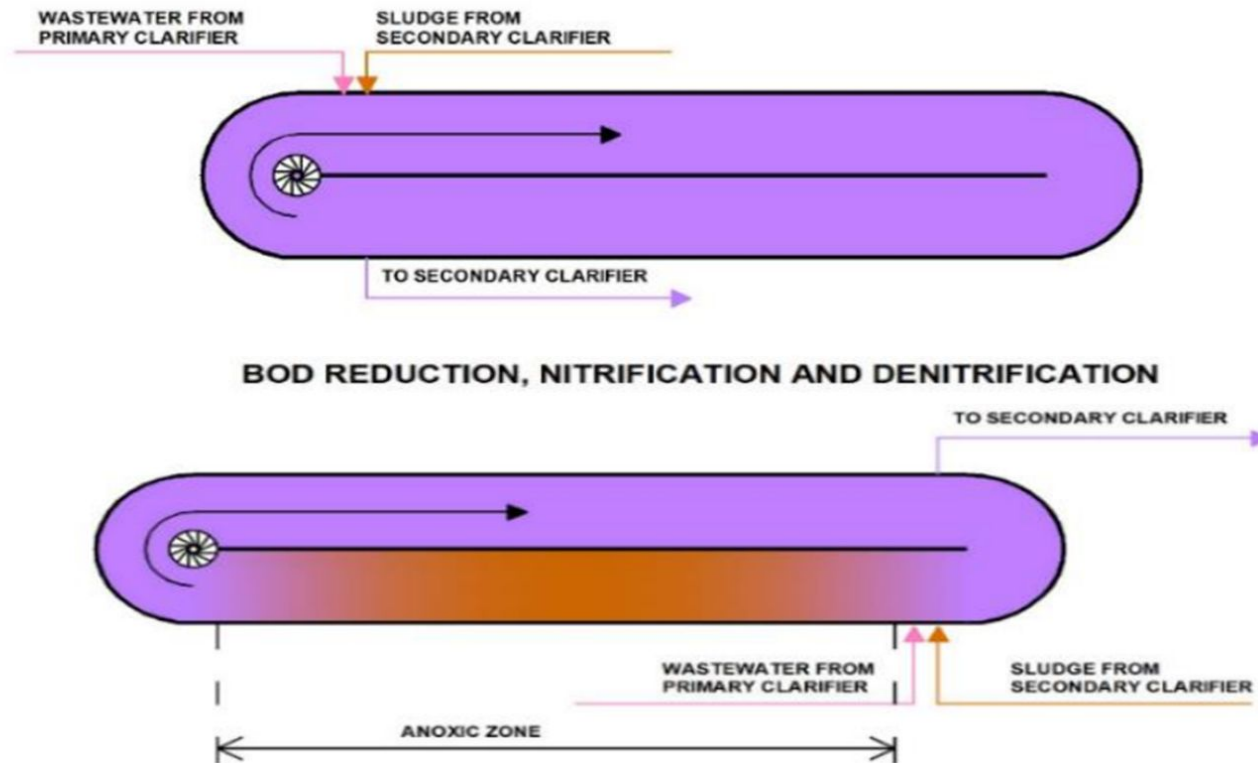
Oxidation Ditch wastewater flow:



Computational simulation of Oxidation ditch:



Oxidation ditch



Schematic diagram of oxidation ditches with BOD removal, nitrification and denitrification

Oxidation ditch: Key design features:

- **Complete Mixing:**

Oxidation ditches are designed for complete mixing, ensuring uniform conditions throughout the channel.

- **Long Solids Retention Time (SRT):**

They operate with a long SRT, which enhances the removal of biodegradable organics and allows for nitrification and denitrification.

- **Flexibility:**

Oxidation ditches can handle variations in wastewater flow and organic loading

- **Nutrient Removal:**

They are capable of achieving good removal of both nitrogen and phosphorus, especially with optimized configurations.

Oxidation ditch: Advantages:

- **Reliable and Stable:**

Known for their ability to handle fluctuations in wastewater flow and composition.

- **Lower Sludge Production:**

Compared to some other biological treatment methods, oxidation ditches produce less sludge.

- **Lower Energy Consumption:**

When designed and operated efficiently, they can be relatively energy-efficient.

- **Ease of Operation and Maintenance:**

Oxidation ditches are generally considered relatively easy to operate and maintain.

Secondary sedimentation Tank/ Clarifier

- A secondary sedimentation tank, also called a secondary clarifier, functions to settle and remove the biological sludge created in the biological treatment process, which contains microorganisms that have broken down organic matter in the wastewater.
- Secondary clarifiers are aimed at separating the biomass and sending it back to the system in the form of RAS for the further processing of incoming wastewater using this settled biomass.
- Produces a clarified effluent that meets discharge standards and concentrated sludge that can be further processed.
- The settled sludge is collected and removed from the tank bottom using a scraper mechanism, while the clarified water overflows at the top.

Outer view of a Secondary Clarifier



An Empty Secondary clarifier



Sludge thickening and dewatering

Sludge thickening is a process in which the solids concentration is increased and the total sludge volume is correspondingly decreased.

Thickening and dewatering start the sludge treatment process in order to:

- **Reduce the volume of the sludge.**
- **Reduce the size of facilities and operating costs.**
- **To optimally separate the water from the matter.**

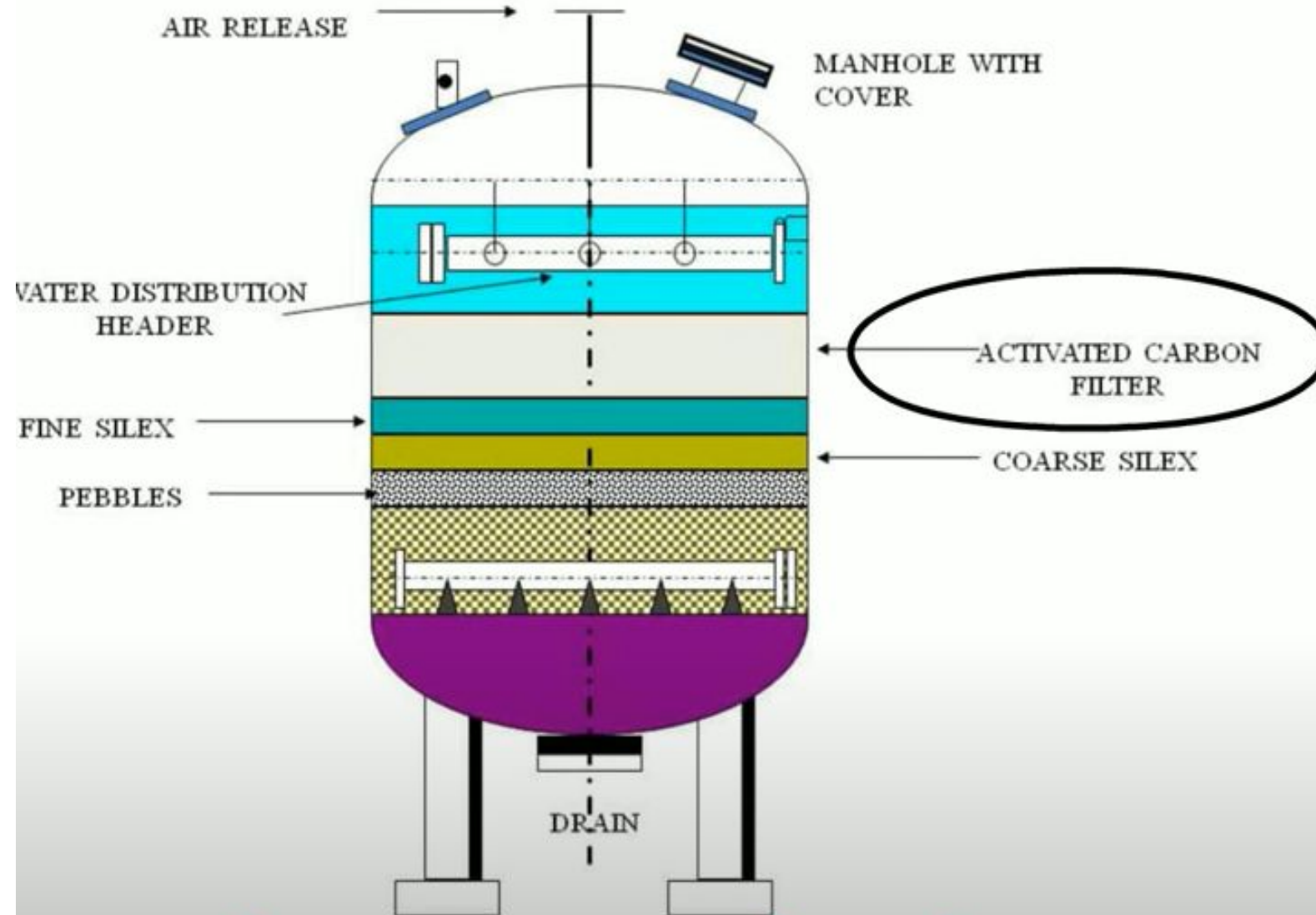
Sludge dewatering machine



Advanced (tertiary) treatment:

Tertiary or advanced wastewater treatment is employed to reduce the residual COD load and when specific wastewater constituents are not removed by the previous treatment stages.

- **Carbon absorption:** Carbon absorption is a wastewater treatment method that uses activated carbon to remove contaminants from water. It's an effective way to remove certain organics, chlorine, fluorine, and radon. It also remove coloring substances and malodor from the water.



Activated Carbon Filtration

Advanced (tertiary) treatment:

- **Ozone Treatment:** Advanced oxidation treatment using ozone is being adopted in some of the CETPs to reduce the non-degradable COD and for color removal. This process has high capital cost and energy requirements.

- **Reverse osmosis (RO):**

Reverse osmosis (RO) is a water treatment process that uses a semi-permeable membrane to purify water by applying pressure, forcing water molecules through the membrane while leaving behind contaminants like salts, minerals, bacteria, and other dissolved solids, effectively producing clean, purified water on one side and leaving the concentrated contaminants on the other side.

Ozonation:

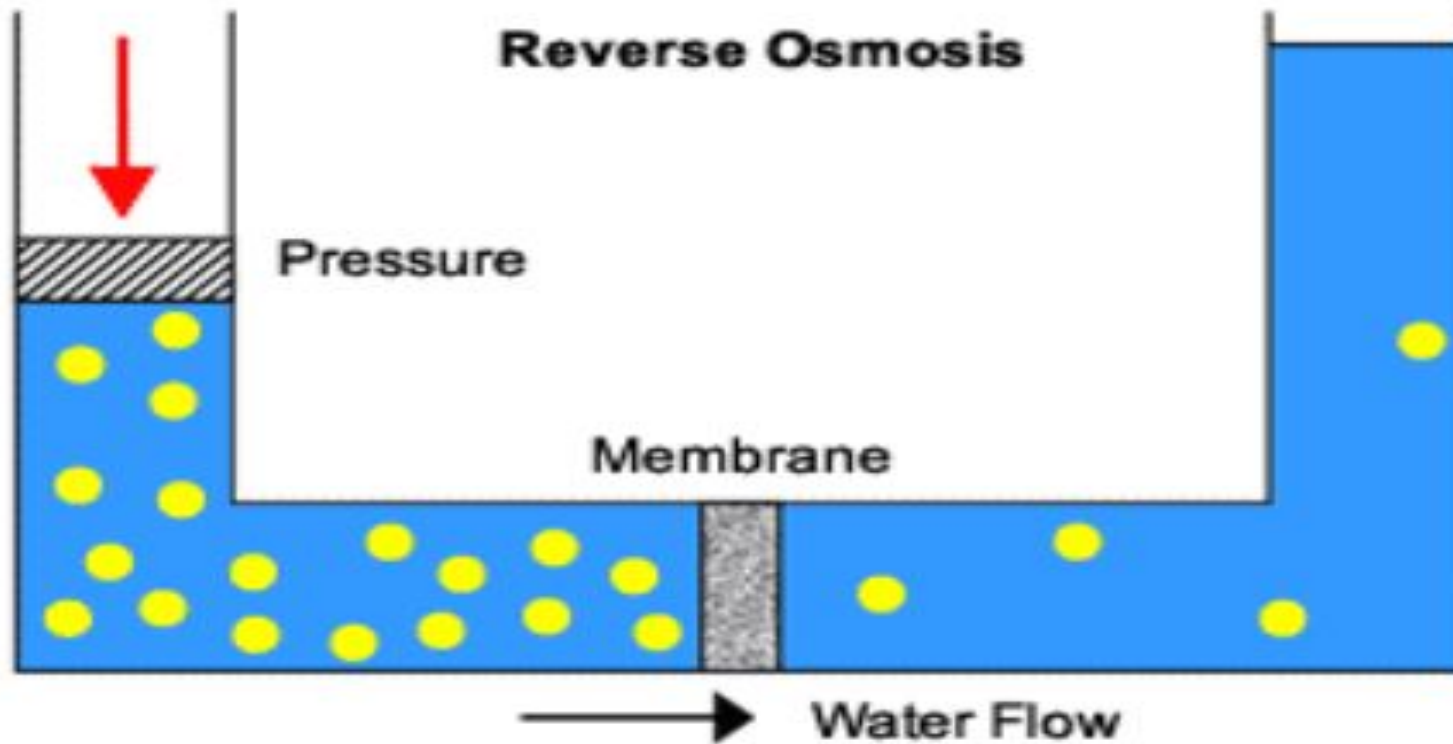


Ozonation

Strong oxidizing agent
Kill microorganisms by destroying
their cell walls



Figure: Reverse osmosis (RO) principle:



The main operational parameter:

- **Total Influent Volume, Q:** Volume of the treated effluent [m³ /day]
- **Tank Volume, V:** Aeration tank volume [m³]
- **Organic Loading, F:** The total BOD₅ applied [kg/day] $F = (f \times Q)/1000$ where f is the BOD₅ of the influent [mg/l]

- **Mixed Liquor Suspended Solids, M:**

$M = (MLSS \times V)/1000$ where MLSS - is the concentration of SS in Mixed Liquor in the aeration tank [mg/l]

The main operational parameter:

- **Hydraulic retention time, T:** $T = (V/Q) \times 24$ where V is the aeration tank volume [m³]
- **Dissolved Oxygen (DO):** the content of molecular oxygen in the aeration tank (mg/l); extended aeration units are usually operated in such a way as to keep the DO of the mixed liquor at about 2-3 mg/l.
- **pH:** Its optimum value is normally between 7.0 and 7.5 with an effective process range of 6-9. The CO₂ produced from respiration neutralizes the excess alkali, adjustment with lime or other needed only if pH drops below 6.

The main operational parameter:

- **F/M (Food to microorganism ratio):**

F/M ratio is defined as the load of substrate (Food) applied per day per unit biomass(bacteria) in the reactor. F/M is usually in the range of 0.2–0.5 kg BOD/kg MLVSS.

A higher F/M ratio means more food is available per microorganism.

- **Surface Hydraulic Loading or Surface Overflow Rate (SOR):** vertical velocity of the influent in the secondary sedimentation tank (m^3/m^2 of tank surface per hour or m/h). SOR of approx. $0.5 \text{ m}^3/\text{m}^2$ per hour is usually adopted for secondary sedimentation of tannery effluents - i.e. less than for primary clarifiers.
- **Surface Solid Rate (SSR):** quantity per hour of MLSS (kg) crossing the surface area of the secondary sedimentation tank (kg/m^2 of tank surface per hour). SSR values between 2.0 and $3.0 \text{ kg}/\text{m}^2$ per hour are usually adopted for secondary sedimentation of tannery effluents.

The main operational parameter:

- **Nutrients:** the nutritional balance of an aerobic system is primarily based upon satisfying the requirements of the cell structure produced by the removal of BOD from waste. Efficient and successful biological oxidation requires a minimal quantity of nitrogen and phosphorus. BOD : N : P ratio of 100 : 5 : 1 in the waste usually ensures adequate nutrition. The tannery effluent is very rich in nitrogen and sometimes poor in phosphorus.
- **Sludge Volume Index (SVI):** is the volume occupied by 1 g of activated sludge after settling the aerated mixed liquor in a 1,000 mL graduated cylinder or Imhoff cone for 30 minutes. It provides good indication of sludge compacting characteristics and it is very helpful in determining return sludge pumping requirements. Well settleable and mineralized sludges have $SVI < 100$.

1. (B) Aerated Lagoon:

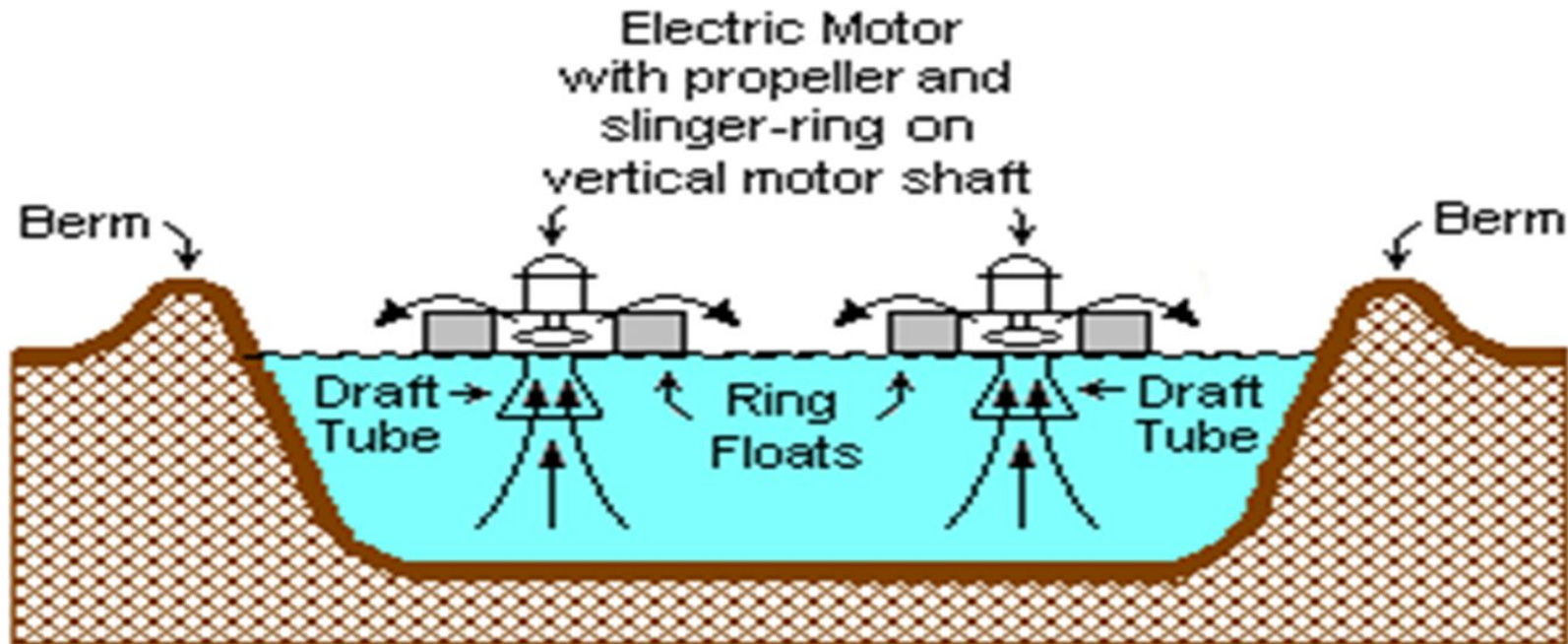
- An **aerated lagoon** (or aerated pond) is a simple wastewater treatment system consisting of a shallow earthen basins from 2 to 6 meter deep with mechanical aerators on floats or fixed platforms. The mechanical aerator are used to provide oxygen for the biological treatment of wastewater and to keep the biological solids in suspension.

Figure: A traditional Aerated Lagoons

AERATED LAGOONS



Figure: Aerated lagoon



A TYPICAL SURFACE – AERATED BASIN

Note: The ring floats are tethered to posts on the berms.

Aerated Lagoons: Advantages

- **Low cost:** Aerated lagoons are inexpensive to build and operate .
- **Simple:** Aerated lagoons are easy to design and operate, making them suitable for rural areas.
- **Natural processes:** Aerated lagoons use natural processes to treat wastewater, reducing the need for chemicals and energy.
- **Habitat for wildlife:** Aerated lagoons can provide a habitat for wildlife and contribute to local biodiversity.
- **Reliable:** Aerated lagoons are reliable for removing BOD.

Aerated Lagoons: Disadvantages

- **Land requirements:** Aerated lagoons require large areas of land, which may not be available in densely populated areas.
- **Slower treatment rate:** Aerated lagoons are slower than other methods of wastewater treatment.
- **Effluent quality:** The quality of the effluent from aerated lagoons may not meet the standards required for certain uses.
- **Odors:** Aerated lagoons can produce odors.
- **Mosquitoes:** Aerated lagoons can attract mosquitoes.

(2) Attached Growth Process:

Rotating biological contactor (RBC):

A rotating biological contactor is a type of biological treatment process. Its working principle is attached growth process. It consists of a series of closely spaced, parallel discs mounted on a rotating shaft which is supported just above the surface of the wastewater. Microorganisms grow on the surface of the discs making a biofilm where biological degradation of the wastewater pollutants takes place.

Figure: Rotating biological contactor (RBC)



RBC: Advantages:

- **High efficiency:** RBCs are very effective at removing organic matter from wastewater
- **Low energy requirements:** RBCs are economical to operate and have low power consumption
- **Simple design:** RBCs are easy to operate and maintain
- **Compact:** RBCs are suitable for densely populated urban areas
- **Good sludge quality:** RBCs produce high quality sludge with a low sludge volume index

RBC: Disadvantages:

- **Clogging:** Clogged discs in the RBC can significantly reduce treatment efficiency .
- **Limited sludge retention:** RBCs may not be suitable for applications that require high sludge retention .
- **Maintenance:** RBCs require regular maintenance and cleaning.
- **High initial cost:** RBCs can have high initial capital costs .
- **Limited applicability:** RBCs may not be suitable for highly contaminated wastewater.

ETP Parameters Limits by DoE, Bangladesh

Sl. No.	Parameters	Units	Typical Values	DOE Standards For Waste from Industrial Units or Project Waste - For Inland Surface Water Discharge
1.	Appearance	-	Colloidal	-
2.	pH	-	8 – 10	6-9
3.	Color	-	Intensively colored	-
4.	Heavy Metals	mg/l	10 – 15	Varies depending on type of metal
5.	Suspended Solids (SS)	mg/l	200 – 300	150
6.	Total Dissolved Solids (TDS)	mg/l	5000 – 6000	2100
7.	Chemical Oxygen Demand (COD)	mg/l	1500 – 1750	200
8.	Bio-chemical Oxygen Demand (BOD)	mg/l	500 – 600	50
9.	Oil & Grease	mg/l	40 – 60	10
10.	Surfactants	mg/l	10 – 40	-
11.	Sulfide as S	mg/l	50 – 60	1

Benefits of ETP:

Fish

Livelihood

☐ Aesthetics

Health

Natural
Resource

Economy



Benefits of Effluent Treatment Plant:

- ETP results in clean, reusable safe water.
- The primary benefit of an Effluent Treatment Plant (ETP) is that it produces clean, reusable water by treating industrial wastewater, allowing industries to recycle and reuse the water instead of constantly drawing fresh water, thus conserving this valuable resource and minimizing environmental impact.

Therefore, treating wastewater by ETP, we can save water and save our planet.

Key benefits of an ETP:

- **Water conservation:**
 - Reduces fresh water usage by recycling treated wastewater for various leather manufacturing processes: in Soaking, Liming and washing.
- **Environmental protection:**
 - Prevents harmful pollutants from being discharged into natural water bodies, safeguarding aquatic ecosystems.
- **Cost savings:**
 - Reduces water bills by minimizing the need for fresh water intake.

Key benefits of an ETP:

- **Protect eutrophication:**
 - By removing nutrients (Nitrogen, Phosphorus and organic carbon) from the effluent, natural water ecosystems can be protected from eutrophication.
- **Pollution control:**
 - Removes hazardous chemicals and contaminants from wastewater, meeting government emission standards.
- **Sustainability:**
 - Promotes a more sustainable approach to water management by minimizing waste and maximizing water reuse.

Thank You